

Senior .NET Core Developer Practical Test

Duration: 3 Days

Stack: .NET Core API, SQL Server, Background Services

Business Scenario

Build a **Retail Sales System** with:

- Inventory Management
 - Sales & Returns
 - Payment Integration (Mock)
 - **Invoice Generation & Print Service**
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Core Modules

1. Inventory API

Endpoints:

- POST /api/items
- PUT /api/stock/update
- GET /api/stock/{sku}

Rules:

- SKU unique (e.g., 105105-25001-RED-SML)
 - Maintain stock ledger
 - Prevent negative stock
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2. Sales API

Endpoints:

- POST /api/sales
- POST /api/sales/{id}/return

- GET /api/sales/{id}

Flow:

- Validate stock
 - Deduct stock
 - Create transaction record
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3. Payment API (Mock NBK)

Simulate:

- POST /api/payment/pay
- POST /api/payment/refund

Requirements:

- Simulate success/failure
 - Store payment status
 - Retry failed transactions
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4. Invoice & Print Service (IMPORTANT)

Requirement

After every successful sale:

1. Generate Invoice
 2. Save Invoice
 3. Send to Print Service
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Invoice Data

- Invoice No
- Date & Time
- Store Name

- Items (Name, Qty, Price)
 - Total Amount
 - Payment Method
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Approach Options

Option A: PDF Invoice

- Generate PDF using:
 - iText7 OR QuestPDF

Option B: Thermal Print (POS Style)

- Generate receipt text (80mm format)
 - Send to printer (simulate)
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Sample Invoice Format

MY STORE

Invoice No: INV-1001

Date: 22-04-2026

Item	Qty	Price
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Shirt	2	10.000
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Shoes	1	25.000
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Total:	45.000
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Payment: CARD

Thank You!